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Carrier Layout Rule Sheet – EEG-CAR-01 Rev C

Document: LAY-EEG-034 **Revision:** A **Date:** 8 September 2026 **Issued by:** TI One Voice research programme (one.witysk.org), Brussels, Belgium **Licence:** CC BY-SA 4.0 **Governing documents:** DSN-EEG-003 Rev D, then RFQ-EEG-001 Rev E, then ICD-EEG-006 Rev C. **Generated by:** tools/emit_rule_sheet.py from tools/rules.py. The tables are not typed.

1. Scope and precedence

This sheet is the rule set for the placement and routing of **EEG-CAR-01 Rev C**, and it is **part of the order specification**. A departure from it is a layout error and is corrected at no charge, not a change request.

tools/design.py governs. Where this sheet and design.py disagree, design.py is right and this sheet is regenerated. Where this sheet and kicad/EEG-CAR-01_RevC.kicad_dru disagree, both are wrong until regenerated: they come from one source, tools/rules.py.

Nothing in this package has been manufactured or measured, and no safety engineer has reviewed this design.

2. Board and stack-up

The full specification is **DSN-EEG-003 Rev D section 3.2** and is not restated. The four lines a layout engineer needs:

- **150.0 x 130.0 mm**, rectangular, no cut-outs, no slots.
- **Four layers:** L1 signal, L2 reference plane, L3 reference plane, L4 signal. 1.60 mm +/- 10 % finished, FR-4 Tg >= 150 C, 1 oz outer and 0.5 oz inner copper, ENIG.
- **Through vias only**, 0.60 mm pad on a 0.30 mm finished hole, tented both sides. No blind, buried, back-drilled, filled or plugged via anywhere.
- Board floor **0.20 mm track / 0.20 mm clearance**, 0.25 mm copper to the board edge, 2.00 mm copper to a non-plated hole. Section 5 is stricter per class and section 5 wins.

3. Zoning and star points

DSN-EEG-003 Rev D section 3.3 rules 1 and 2, cited and not restated. In this sheet's terms:

- The board is two zones split at **x = 62.0 mm**: analogue left, digital right. No digital-class net may be routed into the analogue zone and no analogue-class net out of it. **One signal crosses by design** – CMP_RAW, the comparator output, which must reach a 3.3 V GPIO and does so through R83 and the D23 clamp.
- AGND_REF is the analogue **0 V mid-rail and is not ground**. It is poured on L2 and L3 over the whole analogue zone; DGND is poured over the digital zone on the same two layers. L1 and L4 carry signals, which is why the board has four layers.

- **AGND_REF** joins **DGND** at **R90** and nowhere else. **HARN_SHIELD** joins **DGND** at **R91** and nowhere else. Both are 0 R 0603 parts and neither may become a copper bridge: each has to lift for a leakage measurement and refit. A second path of any kind, on any layer, is a layout error.

4. Isolation keep-out

DSN-EEG-003 Rev D section 3.3 rule 4. The strip $x \geq 141.0$ mm, $y = 2.0$ to 22.0 mm carries **no copper on any of the four layers**, and no via, no pad and no footprint. It is **ISOLATION_KEEPOUT** in the board file and in the .kicad_dru.

The 2.5 kV barrier itself is **on the purchased ADuM4160 module at J10**, not on this board; nothing on the carrier is on the host side. What the carrier owes the barrier is to keep out from under the module's host half. **The rule is enforced against copper EDGES and not track centrelines**, which is what earlier revisions of this programme's own check got wrong.

5. Net classes

Eight classes over all **156** nets. **Minimum** is a floor and a conductor below it is an error; **preferred** is what to route at wherever the board allows it. Vias says whether the class may have one at all. Confirm marks a figure that is a proposal of this programme and not a transcription from a governing document – if one of those is unroutable, say so at the placement gate rather than relaxing it silently.

Class	Nets	Width min / pref (mm)	Clearance min / pref (mm)	Layers	Vias	Confirm
ELECTRODE	32	0.25 / 0.30	0.35 / 0.40	F.Cu	NO	yes
ANALOGUE_REF	5	0.30 / 0.40	0.20 / 0.30	any	yes	yes
POWER	8	0.40 / 0.80	0.20 / 0.30	any	yes	no
USB	2	0.30 / 0.30	0.35 / 0.35	F.Cu	NO	no
ANALOGUE	42	0.25 / 0.25	0.20 / 0.30	any	yes	no
LED_DRIVE	16	0.25 / 0.30	0.20 / 0.25	any	yes	no
DIGITAL	21	0.20 / 0.25	0.20 / 0.25	any	yes	no
DEFAULT	30	0.20 / 0.25	0.20 / 0.25	any	yes	no

Where the numbers come from:

- **ELECTRODE** – Patient-connected. The 0.35 mm clearance is DSN-EEG-003 section 3.3 rule 3 and is not a proposal. The rest of this row IS a proposal: rule 3 says these nets are routed on L1 with the reference plane continuous beneath them, and a via breaks the plane it is routed over, so L1-only and no vias is what rule 3 means read literally. It is achievable – the harness socket J14, the R/D/C protection rows and the module sockets J2/J4/J23/J29 are all on the top side. The 0.25 mm floor is above the board's 0.20 mm because Rev B only reached 0.20 mm on these nets by relaxing, and a patient-connected conductor is the last place to spend the last 0.05 mm. Current is not the constraint: these carry microamperes.

- **ANALOGUE_REF** – AGND_REF, AVDD, AVSS and the module-2 pair. About 10 mA per rail (ICD-EEG-006 section 2.1), so 0.20 mm would carry it four times over: the width is for source impedance and for the mid-rail's noise, not for current. 0.40 mm preferred is what the Rev B router already aimed at.
- **POWER** – VSYS, VBAT, V5V, VBUS_CHG, VBUS_IN, DVDD3V3, VDD_ISO, DGND. The worst case is about 610 mA at J13 (ICD-EEG-006 section 2.7, calculated). On 35 um outer copper at a 10 C rise, IPC-2221 gives about 1.29 A for 0.40 mm and about 0.89 A for 0.25 mm, so 0.40 mm is the floor with margin and 0.80 mm is the preferred width the Rev B router already used.
- **USB** – DSN-EEG-003 section 3.3 rule 5, transcribed and not proposed: a 0.30 mm pair on 0.35 mm spacing on L1 directly over the DGND plane on L2, about 95 ohm differential. The width is the impedance, so it is a fixed value and not a minimum; a via would move the reference and is not permitted.
- **ANALOGUE** – Envelope-detector nodes, the microphone and headphone paths, the comparator. Signal nets at high impedance; 0.25 mm is the board's preferred width and there is no reason for these to be narrower than that.
- **LED_DRIVE** – 1.3 mA per site and 10.4 mA on the LED_V common (ICD-EEG-006 section 2.11, calculated), so current is nowhere near the constraint. These run to a harness connector and out of the box, which is the reason not to route them at the 0.20 mm floor.
- **DIGITAL** – SPI, I2S, I2C, SDMMC, UART and the control lines. Logic-level, milliamperes.
- **DEFAULT** – Everything the classes above do not name: test points, spare ways, no-connects, the harness screen, the button and reset lines.

The class of every net is in kicad/EEG-CAR-01_RevC.kicad_pro and, in a form a person can read, in constraints/nets.md.

6. Geometry rules

These are the seven findings an external layout engineer made against Rev B between 3 and 5 September 2026 (ECO-EEG-016 section 2, ECO-EEG-030), and they are why this sheet exists. **Six are rules; the seventh was a missing check and is section 7.** The right-hand column is not a formality: two of them KiCad cannot check, and this programme checks those itself on the geometry that comes back.

#	Rule	Checked by
1	No drill inside a pad. No via inside an SMD pad or within 0.15 mm of one, on its own net or any other	KiCad DRC (no_via_in_smd_pad) and our DRC on return
2a	No dangling track end. Every track end lands on a pad, a via, another track or a pour	KiCad DRC built-in track_dangling, raised to error in the project file, and our DRC on return
2b	No redundant copper path. Not more than one copper path between the same two points on a net, outside a plane	KiCad DRC built-ins and our DRC on return

#	Rule	Checked by
3	45-degree routing. No angle below 90 degrees between two segments of one track, and every segment on a 0, 45, 90 or 135 degree axis	NOT CHECKABLE IN KiCad – there is no constraint for it. Verified by our DRC on the returned board
4	Pad-centre entry. A track entering a pad has its centreline within 25 % of the pad's minor dimension of the pad centre	NOT CHECKABLE IN KiCad. Verified by our DRC on the returned board
5	Net classes. Section 5 above: width, clearance, permitted layers and the via rule, per class	KiCad DRC (width_*, clearance_*, layers_*, no_vias_*) and our DRC on return
6	Zoning, star points and the isolation keep-out. Sections 3 and 4	KiCad DRC through the rule areas and our DRC on return

kicad/EEG-CAR-01_RevC.kicad_dru is the machine-readable half of this table and carries 37 rules. Load it with the project; it is not optional.

Two more, from DSN-EEG-003 section 3.3, that are not new and are easy to miss.

Rule 7: **no via inside the outline of J2, J4, J23 or J29**, the analogue module connectors – they are NO_VIA_J* rule areas in the board file. Rule 5: **USB_DP and USB_DN are a 0.30 mm pair on 0.35 mm spacing on L1 directly over the L2 DGND plane**, about 95 ohm differential. No coupon and no impedance report are required and none is wanted.

7. Placement

Placement drives via count, must precede routing, and is reviewed before routing starts. That is the reviewer's own principle and this programme has adopted it.

1. **The thirty connectors J1 to J30 are LOCKED**, with MH1-MH4 and FID1-FID3. The pod, the module plate MP-01 and the harness fix them and they may not move. **37 footprints are locked in kicad/EEG-CAR-01_RevC.kicad_pcb and the lock is the specification.** Everything else sits at its Rev B coordinate, unlocked, and is yours to place.
2. **Each of the sixteen protection networks is placed in a straight line – resistor, then clamp, then filter capacitor – running from the harness towards the module** (DSN-EEG-003 section 3.3 rule 3). The schematic is drawn that way so that the intent is visible: sheet 50_harness_input.
3. **Envelope decoupling within 3 mm** of pins 4 and 11 of each OPA4376.
4. **Every electrode-class net is routed on L1 with the reference plane continuous beneath it.** That is what the ELECTRODE class's L1-only, no-via row in section 5 means, and it is DSN-EEG-003 section 3.3 rule 3 read literally.
5. **The placement stage is delivered as a .kicad_pcb with the STEP bodies loaded**, and it passes a 3D collision check with the module bodies installed before it is delivered. The bodies are in mech/step/.

6. **We confirm the placement before routing is confirmed.** The review is a written findings list by net, pad pair and coordinate, capped at four to six hours, under RFQ-EEG-002A as re-scoped by ECO-EEG-030.

One open item that belongs to this programme and not to the layout contractor, stated here because a placement reviewer will meet it: mech/EEG-CAR-01_RevC_collision_check.txt reports that the twelve modules do not fit on MP-01 – 17 356 mm² of declared envelopes into 12 409 mm² of plate – and that the plate's DevKit opening is 2 mm shorter than the DevKit. **Neither is a carrier layout question and neither blocks this order;** both are recorded in ASM-EEG-023 under MECH-D6-MODULE-ENVELOPES.

8. Deliverables expected back

1. **The KiCad project**, at the version agreed at the order – .kicad_pro, .kicad_sch set, .kicad_pcb, and the .kicad_dru unchanged.
2. **A DRC run against this sheet**, clean, taken before handover, with the report. Sections 5 and 6 are the rule set; .kicad_dru is loaded.
3. **Gerber X2**, one file per layer, with file attributes.
4. **Excellon 2 drill files**, plated and non-plated separately.
5. **IPC-D-356A netlist** for the bare-board electrical test.
6. **CPL** for the surface-mount parts and, separately, the through-hole parts.
7. **Fabrication and assembly drawings.**

On return, this programme runs its own DRC over the geometry – including rules 3 and 4 of section 6, which KiCad cannot check – and grades the result. The tool is tools/drc.py with tools/drc_geometry.py, it is in the repository, and it is the same one that regraded Rev B and found 1 311 occurrences of these six rules in a board its own report called clean.

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